



Canned Motor Pump

EtaSeco

Type Series Booklet



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Type Series Booklet EtaSeco

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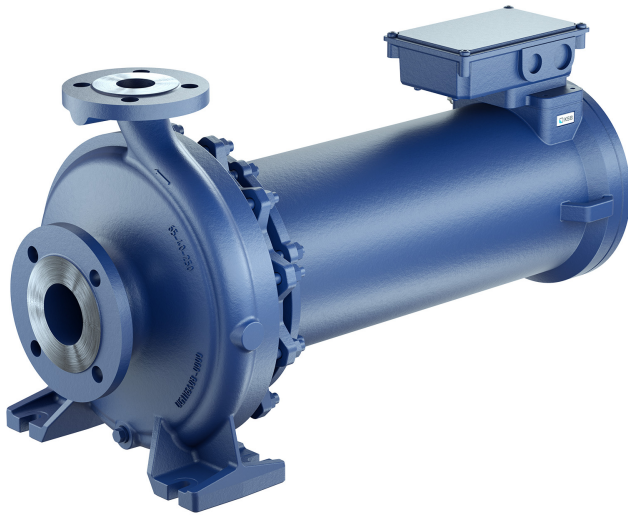
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Seal-less Pumps

Canned Motor Pump

EtaSeco



Main applications

- Cooling circuits
- Transport engineering
- Environmental engineering
- Industry
- Wind power
- Railway

Fluids handled

- Cooling water
- Other coolants
- Volatile fluids
- Valuable fluids

Operating data

Table 1: Operating properties

Characteristic		Value	
		50 Hz	60 Hz
Flow rate	Q [m³/h]	≤ 150	≤ 195
Head	H [m]	≤ 100	≤ 140
Fluid temperature ¹⁾	T [°C]	-50 to 120	-50 to 120
Ambient temperature ¹⁾	T [°C]	-50 to 85	-50 to 85
Nominal pressure	PN [bar]	16	16
Speed	n [rpm]	< 3000	< 3600
Power range of motors	P [kW]	≤ 30	≤ 35

¹ Depending on configuration

² Optional for motor size DS90

Characteristic	Value	
	50 Hz	60 Hz
Electrical connection	Terminal box, bulkhead housing (Harting connector) and cable	
Basic material variant	Nodular cast iron with cathodic electrocoating, nodular cast iron, stainless steel	
Corrosion class	Up to C5 to EN ISO 12944	
Bearing design	Carbon/stainless steel SSiC/SSiC Hybrid combination	
Power supply	360 V, 400 V, 440 V, 480 V, 500 V, 690 V and others on request	
Flanges	DIN, SAE, ASME	
O-rings	EPDM (for water, glycols and other fluids), FKM (for oils and other fluids)	

Design details

Design

- Volute casing pump
- Back pull-out design
- Horizontal installation / vertical installation
- Single-stage

Pump casing

- Radially split volute casing
- Single volute casing (depending on size)
- Main dimensions to EN 733 (flange design to DIN EN or ASME)
- Flanges drilled to DIN EN or ASME or drilled to SAE

Impeller type

- Closed radial impeller with multiply curved vanes
- Axial thrust balancing by means of sealing clearance and balancing holes

Bearings

- Plain bearings
- Product-lubricated

Shaft seal

- Seal-less, with canned motor

Drive

- Three-phase asynchronous motor
- Hermetically sealed canned motor
- Thermal motor protection by PTC thermistors²⁾
- To IEC 60034-7

Motor data
Table 2: Motor data

Parameter	Description
Voltage and frequency	Nominal voltages³⁾: ▪ 50 Hz: 360 V (367 V⁴⁾), 400 V, 500 V, 690 V (3~) ▪ 60 Hz: 400 V, 432 V (440 V⁴⁾), 480 V, 690 V (3~) Voltage tolerances: $\pm 10\%$⁵⁾ Tolerances for voltage ranges $\pm 5\%$, e.g.: 380 - 420 V Frequency tolerance: $\pm 2\%$
Starting method	DOL, star-delta ⁶⁾ or soft starting
Thermal class	H
Operation on a frequency inverter	Possible up to the nominal motor frequency
Enclosure (DIN EN 60034-5)	▪ IP55 – Version with terminal box ▪ IP65 as a minimum⁷⁾ – Version with bulkhead housing, engaged and locked ▪ IP66 – Version with integrated power cable
Duty type (DIN EN 60034-1)	S1, S3
Potential equalisation	Connection for potential equalisation on the outside of the motor housing

-
- 3** Special voltages on request
4 Only for pump models with motor size 12R
5 For 690 V: +5 % and -10 % only
6 Star-delta starting only for pump models with terminal box and motor sizes DS112 and DS132
7 Supplied in IP41 as a minimum
-

Designation

Table 3: Designation example

Position																																			
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
E	S	0		0	5	0	-	0	3	2	-	1	2	5	0	S	G		R	V	P		1	2	S	E	C	C		B					
See name plate and data sheet																														See data sheet					

Table 4: Designation key

Position	Code	Description
1-4	Pump type	
	ESO	EtaSeco
5-15	Size	
	050	Nominal suction nozzle diameter [mm]
	032	Nominal discharge nozzle diameter [mm]
	125	Nominal impeller diameter [mm]
16	Impeller type	
	_{B)}	Hydraulic system optimised for full-load operation in combination with motor 22R/S and ICP motor rating class
	0	Hydraulic system optimised for full-load operation in combination with motor 12R
	1	Hydraulic system optimised for part-load operation in combination with motors 22R/S and ICP motor rating class
	2	Hydraulic system optimised for part-load operation in combination with motor 12R
17	Pump casing material	
	S	Nodular cast iron
		EN-GJS-400-18-LT
		EN-GJS-400-15 / A536 Gr. 60-40-18
	Y	Nodular cast iron + cathodic electrocoating
		EN-GJS-400-18-LT + cathodic electrocoating
		EN-GJS-400-15 / A536 Gr. 60-40-18 + cathodic electrocoating
	C	Stainless steel
		1.4408 / A743 Gr. CF8 M
18	Impeller material	
	G	Cast iron
		EN-GJL-250 / A48 Cl. 35B
		Stainless steel
		1.4408 / A743 Gr. CF8 M
20-22	Motor rating class	
	RVP	Railway vehicle pump
	ICP	Integrated cooling pump
23-24	Motor rating P_N [kW] ⁹⁾	
	1	1,5 kW
	2	2,2 kW
	4	4 kW
	5	5,5 kW
	7	7,5 kW
	11	11 kW
	15	15 kW
	22	22 kW
25	2	Number of motor poles
26	Design	
	S	Standard
	X	Special design
27	Sealing element material (O-rings)	
	E	Special EPDM
	P	Standard EPDM
	V	Standard FKM
	L	Special FKM
	X	Special material
28	Electrical connection	

8 Blank

9 Power output P2 at a specific operating temperature

Position	Code	Description
28	C	Integrated power cable
	P	Bulkhead housing
	T	Terminal box
29	Plain bearing material	
	C	Carbon/stainless steel
	H	Hybrid
	S	SiC/SiC
31	Pump generation	
	B	EtaSeco (generation B)

Table 5: Motor designation key

Motor code	Complete motor designation
12R	DS90.2-1,5R
22R	DS90.2-2,2R
22S	DS90.2-2,2S
42S	DS112.2-4S
52S	DS112.2-5,5S
72S	DS112.2-7,5S
112S	DS132.2-11S
152S	DS132.2-15S
222S	DS132.2-22S

Materials

Table 6: Symbols key

Symbol	Description
x	Standard
o	Optional
-	Version not available / not feasible

Table 7: Overview of available materials

Part No. (⇒ Page 22)	Description	Material	Material variant		
			SG	YG	CC
102	Volute casing	Nodular cast iron EN-GJS-400-18-LT	x ¹⁰⁾	-	-
		Nodular cast iron EN-GJS-400-15 / A536 Gr. 60-40-18	x ¹¹⁾	-	-
		Nodular cast iron + cathodic electrocoating EN-GJS-400-18-LT + cathodic electrocoating	-	x ¹⁰⁾	-
		Nodular cast iron + cathodic electrocoating EN-GJS-400-15/A536 Gr. 60-40-18 + cathodic electrocoating	-	x ¹¹⁾	-
		Stainless steel 1.4408/A743 Gr. CF8M	-	-	x ¹²⁾
230	Impeller	Grey cast iron EN-GJL-250/A48 Cl. 35B	x	x	-
		Stainless steel 1.4408/A743 Gr. CF8M	-	-	x
314	Thrust bearing	Stainless steel 1.4462	x	x	x
		SiC	x	x	x
330	Bearing bracket	Stainless steel 1.4408/A743 Gr. CF8M	x	x	x
400	Gasket	DPAF, asbestos-free	x	x	x
411	Joint ring	DPAF	x	x	x
		PTFE	x	x	x
412	O-ring	EPDM	x	x	x
		FKM	x	x	x
502	Casing wear ring ¹²⁾	JL / lamellar graphite cast iron	x	x	-
529	Bearing sleeve	Stainless steel 1.4462	x	x	x
		SiC	x	x	x
540	Bush ¹²⁾¹³⁾	Stainless steel 1.4021	x	x	x
		Stainless steel 1.4571 ¹¹⁾	o	o	o
545	Bearing bush	Carbon	x	x	x
		SiC	x	x	x
810	Motor part				
	Motor housing	EN-GJL-250 + cathodic electrocoating ¹⁴⁾	x	x	-
		GP240GH+N + cathodic electrocoating ¹⁵⁾	x	x	x
		P235GH ¹¹⁾	o	o	o
		P235GH + cathodic electrocoating ¹¹⁾	x	x	x
	Can	Stainless steel 1.4571 ¹⁴⁾¹⁵⁾	x	x	x
		Stainless steel 1.4539 ¹¹⁾	x	x	x
	Discharge cover ¹⁴⁾¹⁵⁾	Stainless steel 1.4408 / A743 Gr. CF8M ¹⁴⁾	x	x	-
		GP240GH+N + cathodic electrocoating ¹⁵⁾	x	x	x
		Stainless steel 1.4571 ¹⁵⁾	x	x	x
812	Motor housing cover ¹¹⁾	Stainless steel 1.4408/A743 Gr. CF8M	x	x	x
818	Rotor				

10 Not available for motors 112S, 152S, 222S

11 Not available for motors 12R, 22R/S

12 Not available for motor 12R

13 Not available for motors 42S, 52S, 72S with carbon/stainless steel plain bearings

14 Only available for motor 12R

15 Only available for motor 22S

Part No. (⇒ Page 22)	Description	Material	Material variant		
			SG	YG	CC
818	Can	Stainless steel 1.4539 ¹⁴⁾¹⁵⁾	X	X	X
		Stainless steel 1.4571 ¹¹⁾	X	X	X
	Shaft	Stainless steel 1.4021	X	X	X
		Stainless steel 1.4462	o	o	o
902	Studs	Steel 8.8 ¹²⁾	X	X	-
		A2-70 ¹⁴⁾	X	X	-
		A4-70 / A193 Gr. B8M Cl. 2 ¹²⁾	o	o	X
903	Screw plugs	Steel	X	-	-
		A4/AISI 316	o	X	X

Coating and preservation

- Coating and preservation to KSB standard

Product benefits

- **Reliability and dependability**
 - Hermetically sealed canned motor pump with a long service life of up to 30 years and very long maintenance intervals for minimal life cycle costs
 - Seal-less, leakage-free design for maximum system availability, with excellent MTBF and no unscheduled downtimes
 - Low-noise operation and smooth running for stable and quiet performance
- **Long service life and efficiency**
 - Robust, corrosion-resistant design up to corrosion class C5 (EN ISO 12944), available in stainless steel or nodular cast iron with cathodic electrocoating
 - Motors developed and manufactured by KSB, reliable heavy-duty plain bearings made of SSiC or carbon for maximum service life
 - Suitable for operation on a frequency inverter, combined with energy-efficient hydraulic components
- **Flexible and compact**
 - Easy to install thanks to low-weight and compact design for horizontal and vertical installation
 - 8 motor sizes for different voltages and 3 connection types available: terminal box, surface-mounted housing or cable
 - 24 hydraulic sizes with mating dimensions to EN 733 and flanges to DIN / drilled to SAE and ASME

Product information

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

Acceptance tests and warranty

- Materials inspection and testing
 - Test report 2.2 on request
- Final inspection
 - Inspection certificate 3.1 to EN 10204 on request
- Hydraulic test

The duty point of each pump is guaranteed to ISO 9906/3B.

The following acceptance tests can be performed and certified at extra charge:

- Performance test to ISO 9906¹⁶⁾

- NPSH test
- Other inspections/tests on request
- Warranty

Warranties are given within the scope of the valid terms and conditions of sale and delivery.

Overview of fluids handled

Table 8: Symbols key

Symbol	Description
X	Standard
-	Version not available / not feasible

Table 9: Overview of fluids handled with associated material variants

Fluid handled ¹⁷⁾¹⁸⁾	Application limits		Material variant		
	Max. content	T _{max}	SG	YG	CC
	%	°C			
Alkaline solution (with 2 % NaOH)	-	-	X	-	X
Ammonium hydroxide	≤ 25	≤ +30	X	-	X
Ethylene glycol base anti-freeze, inhibited, closed system	≤ 50	-	X	X	X
Ethylene glycol base anti-freeze, inhibited, closed system: Antifrogen N	≤ 60	-	X	X	X
Propylene glycol base anti-freeze, inhibited, closed system	≤ 50	-	X	X	X
Cooling brine on potassium carbonate basis	≤ 50	≤ +30	-	-	X
Cooling brines on alkaline earth formate or acetate basis (e.g. Tyfoxit, Pekasol, Glysofor)	-	-	-	-	X
Sodium carbonate	≤ 10	≤ +30	-	-	X
Lubricating oil/hydraulic oil: Shell TELLUS 46	-	-	X	X	X
Turbine oil ¹⁹⁾	-	-	X	X	X
Thermal oil MIDEAL 7131	-	-	X	X	X
Thermal oil Therminol D12	-	-	X	X	X
Water ²⁰⁾					
▪ Cooling water (closed cooling circuit)	-	≤ +60 ²¹⁾	X	X	X
▪ Clean water	-	≤ +70	X	X	X
▪ Desalinated water ²²⁾	-	-	-	-	X
▪ Ultra-pure water ²²⁾	-	-	-	-	X

16 For ISO 9906/2B performance guarantee instead of efficiency guarantee.

17 O-ring material
FKM: Permissible down to -30 °C
EPDM: Permissible down to -40 °C

18 The carbon bearing can only be used for fluids handled with a low solids content (< 20 ppm) and a low particle hardness (Mohs hardness ≤ 4).

19 Not applicable to SFD oils, hardly flammable

20 General evaluation criteria for results of water analysis: pH > ≥ 6.5; chlorides content (Cl) ≤ 150 mg/kg, chlorine (Cl₂) ≤ 0.6 mg/kg If limits are exceeded contact KSB Service.

21 Temperatures > 60 °C increase the risk of limescale deposits. Only permissible under special conditions (softening)

22 For desalinated and ultra-pure water operating limits regarding conductivity apply. Consult with KSB Service.

Motor versions

Table 10: Symbols key

Symbol	Description
X	Standard
-	Version not available / not feasible

Table 11: Material variant and motor size per pump size

Size	Motor											
	12R			22R/S			42S, 52S, 72S			112S, 152S, 222S		
	Material variant											
	SG	YG	CC	SG	YG	CC	SG	YG	CC	SG	YG	CC
032-032-160 ²³⁾	-	-	-	-	X	-	-	-	-	-	-	-
040-025-160	-	-	-	-	X	X	-	-	-	-	-	-
040-025-200	-	-	-	-	-	-	X	X	X	-	-	-
050-032-125	-	-	-	-	X	X	-	-	-	-	-	-
050-032-125.0	X	X	-	-	-	-	-	-	-	-	-	-
050-032-125.1	-	-	-	-	X	X	-	-	-	-	-	-
050-032-125.2	X	X	-	-	-	-	-	-	-	-	-	-
050-032-160	-	-	-	X	X	X	X	X	X	-	-	-
050-032-160.1	-	-	-	X	X	X	X	X	X	-	-	-
050-032-200	-	-	-	-	-	-	X	X	X	X	X	X
050-032-200.1	-	-	-	-	-	-	X	X	X	X	X	X
050-032-250	-	-	-	-	-	-	-	-	-	X	X	X
050-032-250.1	-	-	-	-	-	-	-	-	-	X	X	X
065-040-125	-	-	-	X	X	X	X	X	X	-	-	-
065-040-160	-	-	-	-	-	-	X	X	X	-	-	-
065-040-200	-	-	-	-	-	-	X	X	X	X	X	X
065-040-250	-	-	-	-	-	-	-	-	-	X	X	X
065-050-125	-	-	-	-	-	-	X	X	X	-	-	-
065-050-160	-	-	-	-	-	-	X	X	X	-	-	-
065-050-200	-	-	-	-	-	-	X	X	X	X	X	X
065-050-250	-	-	-	-	-	-	-	-	-	X	X	X
080-065-125	-	-	-	-	-	-	X	X	X	-	-	-
080-065-160	-	-	-	-	-	-	X	X	X	-	-	-
080-065-200	-	-	-	-	-	-	-	-	-	X	X	X

Pressure limits and temperature limits

Table 12: Pressure limits and fluid temperature

Material variant	Plain bearing, pump end	Motor							
		12R		22R		22S		42S, 52S, 72S, 112S, 152S, 222S	
		PN10		PN10		PN16		PN16	
		T _{min}	T _{max}	T _{min}	T _{max} ²⁴⁾	T _{min}	T _{max} ²⁴⁾	T _{min}	T _{max}
		°C							
SG, YG ²⁵⁾	Carbon	-40	95	-40	95	-40	95	-40	120
SG, YG ²⁵⁾	SiC	-	-	-40	95	-40	95	-40	120
CC	Carbon	-	-	-40 ²⁶⁾	95	-40 ²⁶⁾	95	-40	120
CC	SiC	-	-	-40 ²⁶⁾	95	-40 ²⁶⁾	95	-40	120

23 In-line pump

24 Version with terminal box permissible up to T_{max} = 110 °C

25 Combustible fluids (except MIDEAL 7131) only permissible up to T_{min} = -10 °C

26 With suitable O-ring material possible down to -50 °C. Consult KSB.

Noise characteristics

Table 13: Surface sound pressure level, three-phase motor

Motor code	Motor size	Surface sound pressure level ⁽²⁷⁾⁽²⁸⁾
		L_{pA}
		[dB]
12R	DS90.2-1,5R	49
22R/22S	DS90.2-2,2R/S	49
42S	DS112.2-4S	52
52S	DS112.2-5,5S	53
72S	DS112.2-7,5S	55
112S	DS132.2-11S	58
152S	DS132.2-15S	60
222S	DS132.2-22S	62

Electrical connection

Table 14: Symbols key

Symbol	Description
X	Standard
-	Version not available / not feasible

Table 15: Overview of electrical connection per motor size

Part No.	Description	Enclosure	Motor							
			12R	22R/S	42S	52S	72S	112S	152S	222S
833	Terminal box									
	Square	IP 55	-	X	X	X	X	-	-	-
	Rectangular	IP 55	-	-	-	-	-	X	X	X
81-99	Bulkhead housing									
	3HPR									
	▪ Crimp insert Q3/O HV	IP 66	X	X	X	X	-	-	-	-
	▪ Crimp insert Q5/O	IP 66	X	X	-	-	-	-	-	-
	6B									
	▪ Crimp insert 6E	IP 65	X	X	X	-	-	-	-	-
	▪ Crimp insert 10EE	IP 65	X	X	X	-	-	-	-	-
	6HPR	IP 66	-	-	X	X	X	X	X	-
	10HPR	IP 66	-	-	-	-	-	X	X	X
	16HPR	IP 66	-	-	-	-	-	-	-	X
824	Power cable									
	Angle gland	IP 66	X	X	-	-	-	-	-	-
	Cable gland	IP 66	-	-	X	X	X	X	-	-

27 Surface sound pressure level to ISO 3744 and EN 12639; valid for a pump operating range of $Q/Q_{BEP} = 0.8$ to 1.1 and non-cavitating operation. If noise levels are to be guaranteed, add + 3 dB for measuring and constructional tolerance.

28 Increase for 60 Hz operation: 3500 rpm + 3dB, 1750 rpm + 1 dB

Technical data

EtaSeco

Table 16: Technical data

Size	Number of vanes	Impeller outlet width	Impeller inlet diameter	Nominal impeller diameter		Weight of pump set with motor ²⁹⁾							
				Maximum	Minimum	12R ³⁰⁾	22R/S	42S	52S	72S	112S	152S	222S
		[mm]				[kg]							
032-032-160	6	5,7	52,7	170	110	-	45	-	-	-	-	-	-
040-025-160	4	6,0	45,2	169	94	-	45	-	-	-	-	-	-
040-025-200	4	6,0	45,2	209	165	-	-	-	81	-	-	-	-
050-032-125	6	9,8	63,4	139	94	-	44	-	-	-	-	-	-
050-032-125.0	6	9,8	63,2	139	94	27,5	-	-	-	-	-	-	-
050-032-125.1	6	6,8	52,5	139	94	-	44	-	-	-	-	-	-
050-032-125.2	6	6,8	52,0	139	104	27,5	-	-	-	-	-	-	-
050-032-160	6	8,5	60,6	174	120	-	45	64	-	-	-	-	-
050-032-160.1	6	5,7	52,7	170	120	-	45	-	-	-	-	-	-
050-032-200	6	7,0	62,9	209	170	-	-	-	-	97	-	-	-
050-032-200.1	6	5,6	54,0	204	168	-	-	-	-	97	-	-	-
050-032-250	6	7,5	62,6	261	205	-	-	-	-	-	121	-	-
050-032-250.1	6	5,5	58,3	254	200	-	-	-	-	-	-	-	155
065-040-125	6	14,0	73,9	139	92	-	47	67	-	-	-	-	-
065-040-160	6	13,0	70,0	174	128	-	-	-	74	91	-	-	-
065-040-200	6	9,4	69,4	209	160	-	-	-	-	98	-	-	145
065-040-250	6	8,4	74,1	260	195	-	-	-	-	-	123	-	157
065-050-125	6	19,9	87,9	142	112	-	-	-	77	-	-	-	-
065-050-160	6	16,9	86,9	174	128	-	-	-	88	-	-	-	-
065-050-200	6	13,8	83,6	219	170	-	-	-	-	101	110	-	-
065-050-250	6	10,5	84,0	260	215	-	-	-	-	-	123	-	-
080-065-125	6	25,8	99,0	141	109	-	-	-	-	98	-	-	-
080-065-160	6	21,0	91,9	174	132	-	-	76	85	-	-	-	-
080-065-200	6	17,0	99,7	219	148	-	-	-	-	-	-	133	155

29 Motors 22R/S-222S: Weight indicated for pump with EN flange without support foot, in material variant SG. Addition for material variant CC: approx. + 5 kg. Addition for support foot: approx. + 0.4 kg

30 Motor 12R: Weight indicated for pump with SAE flange without support foot, in material variant SG. Addition for EN flange: approx. + 2 kg. Addition for support foot: approx. + 2.5 kg

Motors 50 Hz
Table 17: Technical data of motors: f = 50 Hz (+/- 1 %); V_N = 400 V, 500 V (+/- 10 %), 690 V (+ 5/- 10 %)

Motor	Motor size	Fluid temperature	Operating data at rated power								Starting current	Starting torque	Moment of inertia	Weight of complete drive
			Rated current			Speed	Rated power	Efficiency	Power factor	Torque				
			I _N											
			400 V	500 V	690 V									
T				n _N	P ₂	η	cos φ	M _N ³¹⁾	I _N ³²⁾	M _N ³²⁾	J			
[°C]	[A]			[rpm]	[kW]	[%]		[Nm]	[A]	[Nm]	[kgm ²]	[kg]		
12R	DS90.2-1,5R	40	4,7	3,8	3,3	2720	2,1	74,5	0,83	7,4	4,2	1,9	0,003	35
12R	DS90.2-1,5R	60	4,7	3,8	3,3	2720	2,0	74,5	0,83	7,4	4,2	1,9	0,003	35
12R	DS90.2-1,5R	80	4,2	3,4	3,0	2760	1,9	75,3	0,81	7,4	4,7	1,9	0,003	35
12R	DS90.2-1,5R	100	3,8	3,1	2,7	2800	1,7	75,6	0,77	7,4	5,2	1,9	0,003	35
12R	DS90.2-1,5R	120	3,4	2,8	2,4	2830	1,4	75,4	0,73	7,4	5,7	1,9	0,003	35
22R/S	DS90.2-2,2R/S	40	7,6	6,2	4,5	2785	3,2	74,6	0,86	11,0	4,2	2,0	0,003	35
22R/S	DS90.2-2,2R/S	60	7,2	5,8	4,2	2810	3,1	75,3	0,86	11,0	4,5	2,0	0,003	35
22R/S	DS90.2-2,2R/S	80	6,8	5,5	4,0	2810	2,8	75,6	0,84	11,0	4,7	2,0	0,003	35
22R/S	DS90.2-2,2R/S	100	6,0	4,8	3,5	2845	2,5	75,5	0,82	11,0	5,4	2,0	0,003	35
22R/S	DS90.2-2,2R/S	120	5,0	4,0	2,9	2920	1,9	74,4	0,74	11,0	6,5	2,0	0,003	35
42S	DS112.2-4S	40	13,8	10,8	7,4	2820	6,3	76,1	0,88	21,5	3,8	1,7	0,006	57
42S	DS112.2-4S	60	12,9	10,2	6,9	2840	6,0	76,6	0,88	21,5	4,1	1,7	0,006	57
42S	DS112.2-4S	80	12,1	9,6	6,5	2850	5,5	77,0	0,87	21,5	4,4	1,7	0,006	57
42S	DS112.2-4S	100	10,9	8,7	6,0	2870	5,0	77,3	0,86	21,5	4,9	1,7	0,006	57
42S	DS112.2-4S	120	9,4	7,7	5,2	2900	4,3	77,3	0,83	21,5	5,6	1,7	0,006	57
52S	DS112.2-5,5S	40	18,8	14,3	10,4	2830	9,0	80,4	0,90	30,3	4,6	2,8	0,008	62
52S	DS112.2-5,5S	60	17,7	13,5	9,8	2850	8,4	81,0	0,89	30,3	4,8	2,8	0,008	62
52S	DS112.2-5,5S	100	13,4	10,2	7,4	2900	6,0	79,7	0,84	30,3	6,4	2,8	0,008	62
52S	DS112.2-5,5S	100	13,4	10,2	7,4	2900	6,0	79,7	0,84	30,3	6,4	2,8	0,008	62
52S	DS112.2-5,5S	120	9,8	7,6	5,5	2940	3,8	78,9	0,74	30,3	8,7	2,8	0,008	62
72S	DS112.2-7,5S	40	29,4	22,3	16,0	2860	13,9	79,7	0,87	46,4	5,7	1,7	0,021	72
72S	DS112.2-7,5S	60	26,5	20,1	14,6	2870	12,7	80,2	0,86	46,4	6,3	1,7	0,021	72
72S	DS112.2-7,5S	80	24,5	18,7	13,6	2890	11,6	80,3	0,86	46,4	6,8	1,7	0,021	72
72S	DS112.2-7,5S	100	22,4	17,2	12,6	2900	10,4	80,1	0,84	46,4	7,5	1,7	0,021	72
72S	DS112.2-7,5S	120	18,6	14,4	10,6	2930	8,0	79,2	0,79	46,4	9,0	1,7	0,021	72
112S	DS132.2-11S	40	33,2	25,1	19,9	2880	17,2	80,9	0,93	57,1	6,2	2,5	0,026	72
112S	DS132.2-11S	60	31,0	23,6	18,3	2890	16,0	81,0	0,93	57,1	6,6	2,5	0,026	72
112S	DS132.2-11S	80	28,5	21,8	16,6	2900	14,6	81,0	0,92	57,1	7,2	2,5	0,026	72
112S	DS132.2-11S	100	26,1	20,0	15,2	2900	13,1	80,9	0,91	57,1	7,9	2,5	0,026	72
112S	DS132.2-11S	120	23,1	17,8	13,4	2920	11,4	80,5	0,90	57,1	8,9	2,5	0,026	72
152S	DS132.2-15S	40	39,8	28,6	22,0	2890	20,6	81,1	0,92	68,1	6,0	2,4	0,030	85
152S	DS132.2-15S	60	37,2	26,9	20,7	2900	19,2	81,3	0,92	68,1	6,4	2,4	0,030	85
152S	DS132.2-15S	80	33,6	24,5	18,8	2910	17,2	81,1	0,91	68,1	7,1	2,4	0,030	85
152S	DS132.2-15S	100	30,2	22,1	17,0	2920	15,3	80,7	0,90	68,1	7,8	2,4	0,030	85
152S	DS132.2-15S	120	26,3	19,3	14,8	2930	12,8	79,6	0,89	68,1	9,0	2,4	0,030	85
222S	DS132.2-22S	40	54,2	42,6	30,2	2890	28,5	81,9	0,93	94,1	4,6	2,4	0,040	112
222S	DS132.2-22S	60	49,9	39,5	27,9	2900	26,0	81,9	0,92	94,1	5,0	2,4	0,040	112
222S	DS132.2-22S	80	45,4	36,1	25,5	2910	23,4	81,6	0,92	94,1	5,5	2,4	0,040	112
222S	DS132.2-22S	100	41,1	32,9	23,2	2920	20,8	81,1	0,91	94,1	6,1	2,4	0,040	112
222S	DS132.2-22S	120	36,0	29,0	20,4	2940	17,6	80,1	0,89	94,1	6,9	2,4	0,040	112

31 Torque as maximum value at a fluid temperature of 40 °C

32 For DOL starting a multiple thereof

Motors 60 Hz
Table 18: Technical data of motors: $f = 60 \text{ Hz}$ (+/- 1 %); $V_N = 400 \text{ V}$, 480 V (+/- 10 %), 690 V (+ 5/- 10 %)

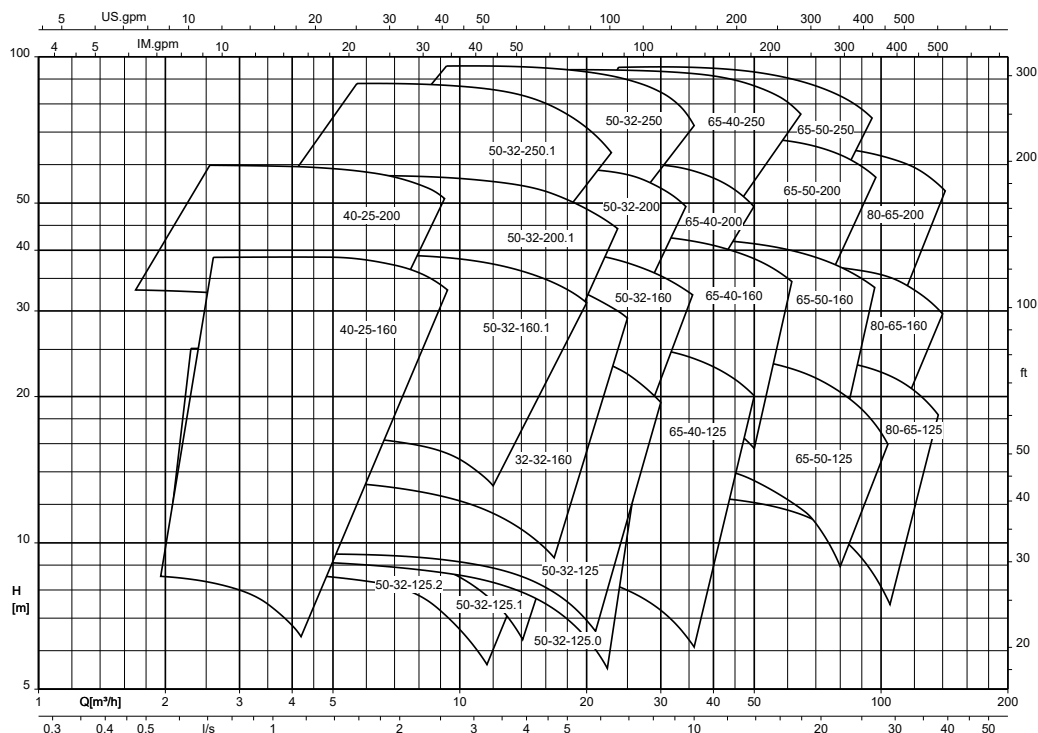
Motor	Motor size	Fluid temperature	Operating data at rated power								Starting current	Starting torque	Moment of inertia	Weight of complete drive
			Rated current			Speed	Rated power	Efficiency	Power factor	Torque				
			T	I _N			n _N	P ₂	η	cos φ				
[°C]	[A]			[rpm]	[kW]	[%]		[Nm]	[A]	[Nm]	[kgm ²]	[kg]		
12R	DS90.2-1,5R	40	4,5	3,6	2,6	3420	2,5	73,9	0,77	7,0	4,2	1,9	0,003	35
12R	DS90.2-1,5R	60	4,5	3,6	2,6	3420	2,4	73,9	0,77	7,0	4,3	1,9	0,003	35
12R	DS90.2-1,5R	80	4,2	3,4	2,4	3440	2,2	73,3	0,74	7,0	4,7	1,9	0,003	35
12R	DS90.2-1,5R	100	4,2	3,4	2,4	3440	2,0	73,3	0,74	7,0	4,7	1,9	0,003	35
12R	DS90.2-1,5R	120	3,6	2,9	2,1	3480	1,7	71,5	0,67	7,0	5,4	1,9	0,003	35
22R/S	DS90.2-2,2R/S	40	8,0	6,6	4,7	3360	3,9	76,8	0,85	11,1	4,2	2,0	0,003	35
22R/S	DS90.2-2,2R/S	60	7,7	6,3	4,4	3370	3,7	76,8	0,84	11,1	4,4	2,0	0,003	35
22R/S	DS90.2-2,2R/S	80	7,3	6,0	4,2	3390	3,4	76,8	0,83	11,1	4,6	2,0	0,003	35
22R/S	DS90.2-2,2R/S	100	6,6	5,5	3,8	3420	3,0	76,3	0,80	11,1	5,1	2,0	0,003	35
22R/S	DS90.2-2,2R/S	120	5,7	4,7	3,3	3460	2,3	75,3	0,74	11,1	6,0	2,0	0,003	35
42S	DS112.2-4S	40	13,8	11,1	7,5	3460	6,3	76,5	0,85	17,4	3,8	1,7	0,006	57
42S	DS112.2-4S	60	13,1	10,5	7,2	3470	5,9	76,5	0,84	17,4	4,1	1,7	0,006	57
42S	DS112.2-4S	80	12,3	9,9	6,8	3470	5,4	76,3	0,83	17,4	4,3	1,7	0,006	57
42S	DS112.2-4S	100	11,3	9,0	6,3	3480	4,8	75,7	0,81	17,4	4,7	1,7	0,006	57
42S	DS112.2-4S	120	10,0	8,0	5,6	3500	4,1	74,1	0,77	17,4	5,3	1,7	0,006	57
52S	DS112.2-5,5S	40	18,5	15,6	10,6	3440	9,1	79,1	0,87	25,1	4,6	2,8	0,008	62
52S	DS112.2-5,5S	60	17,4	14,7	10,0	3460	8,4	79,3	0,87	25,1	4,9	2,8	0,008	62
52S	DS112.2-5,5S	100	15,4	12,9	8,8	3480	7,1	78,8	0,84	25,1	5,6	2,8	0,008	62
52S	DS112.2-5,5S	100	15,4	12,9	8,8	3480	7,1	78,8	0,84	25,1	5,6	2,8	0,008	62
52S	DS112.2-5,5S	120	13,5	11,3	7,7	3510	5,9	77,4	0,81	25,1	6,3	2,8	0,008	62
72S	DS112.2-7,5S	40	28,8	16,2	24,3	3460	13,7	80,5	0,85	38,0	5,8	1,7	0,021	72
72S	DS112.2-7,5S	60	26,5	15,0	22,3	3480	12,4	80,2	0,83	38,0	6,3	1,7	0,021	72
72S	DS112.2-7,5S	80	24,9	14,1	21,0	3490	11,3	79,8	0,82	38,0	6,7	1,7	0,021	72
72S	DS112.2-7,5S	100	23,1	13,1	19,5	3500	10,1	79,0	0,80	38,0	7,3	1,7	0,021	72
72S	DS112.2-7,5S	120	19,5	11,1	16,5	3530	7,7	76,1	0,73	38,0	8,6	1,7	0,021	72
112S	DS132.2-11S	40	33,9	26,3	27,5	3470	16,2	78,1	0,91	44,6	6,1	2,5	0,026	72
112S	DS132.2-11S	60	31,7	24,7	25,7	3480	14,9	77,7	0,90	44,6	6,5	2,5	0,026	72
112S	DS132.2-11S	80	29,3	22,8	23,7	3490	13,5	76,9	0,89	44,6	7,0	2,5	0,026	72
112S	DS132.2-11S	100	26,6	20,7	21,5	3510	11,9	75,8	0,88	44,6	7,7	2,5	0,026	72
112S	DS132.2-11S	120	23,8	18,5	19,2	3520	10,2	73,8	0,86	44,6	8,6	2,5	0,026	72
152S	DS132.2-15S	40	41,3	33,6	22,8	3490	20,5	80,0	0,91	56,1	5,7	2,4	0,030	85
152S	DS132.2-15S	60	38,9	31,6	21,5	3490	19,0	79,6	0,90	56,1	6,1	2,4	0,030	85
152S	DS132.2-15S	80	35,5	28,9	19,7	3510	17,1	78,8	0,89	56,1	6,7	2,4	0,030	85
152S	DS132.2-15S	100	32,6	26,5	18,2	3510	15,1	77,8	0,88	56,1	7,3	2,4	0,030	85
152S	DS132.2-15S	120	28,8	23,4	16,2	3530	12,7	75,8	0,86	56,1	8,2	2,4	0,030	85
222S	DS132.2-22S	40	54,9	45,2	31,2	3480	27,6	78,9	0,91	75,8	4,6	2,4	0,040	112
222S	DS132.2-22S	60	50,7	41,7	28,9	3490	25,0	78,2	0,90	75,8	4,9	2,4	0,040	112
222S	DS132.2-22S	80	46,6	38,3	26,6	3500	22,4	77,2	0,89	75,8	5,4	2,4	0,040	112
222S	DS132.2-22S	100	42,7	35,0	24,4	3510	19,8	75,8	0,88	75,8	5,9	2,4	0,040	112
222S	DS132.2-22S	120	37,6	30,8	21,6	3530	16,5	73,3	0,85	75,8	6,7	2,4	0,040	112

33 Torque as maximum value at a fluid temperature of 40 °C

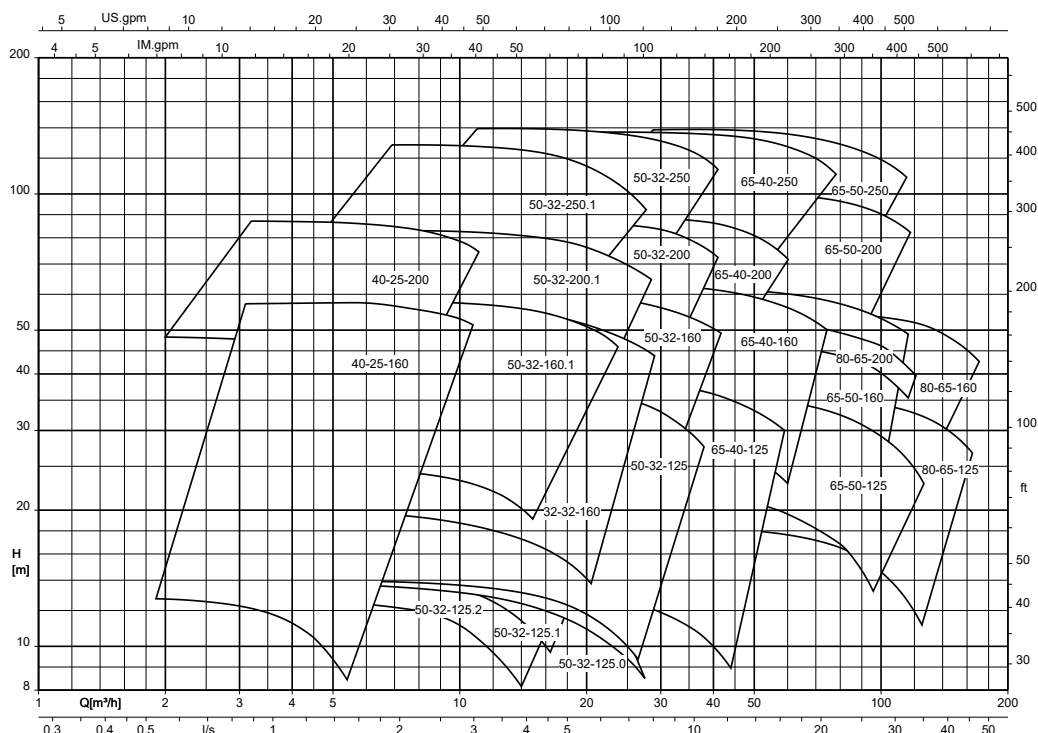
34 For DOL starting a multiple thereof

Selection charts

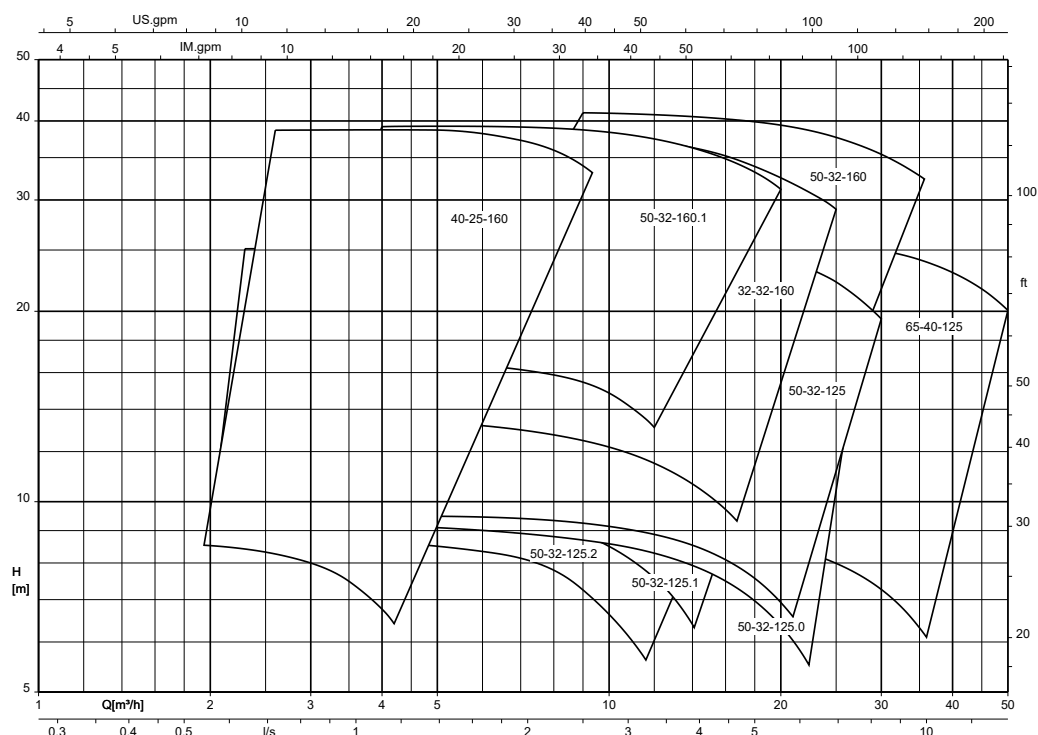
EtaSeco, 2900 rpm



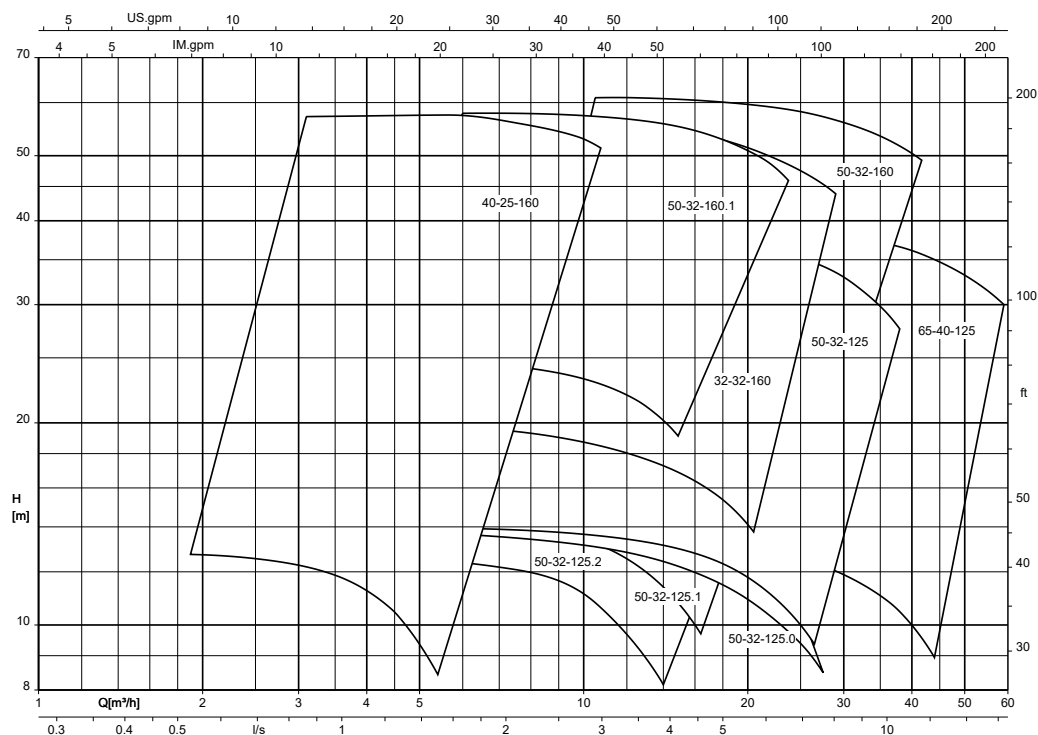
EtaSeco, 3500 rpm



EtaSeco motors 12 and 22, 2900 rpm



EtaSeco motors 12 and 22, 3500 rpm



Dimensions

Motor size DS90, motor sizes 22R/S

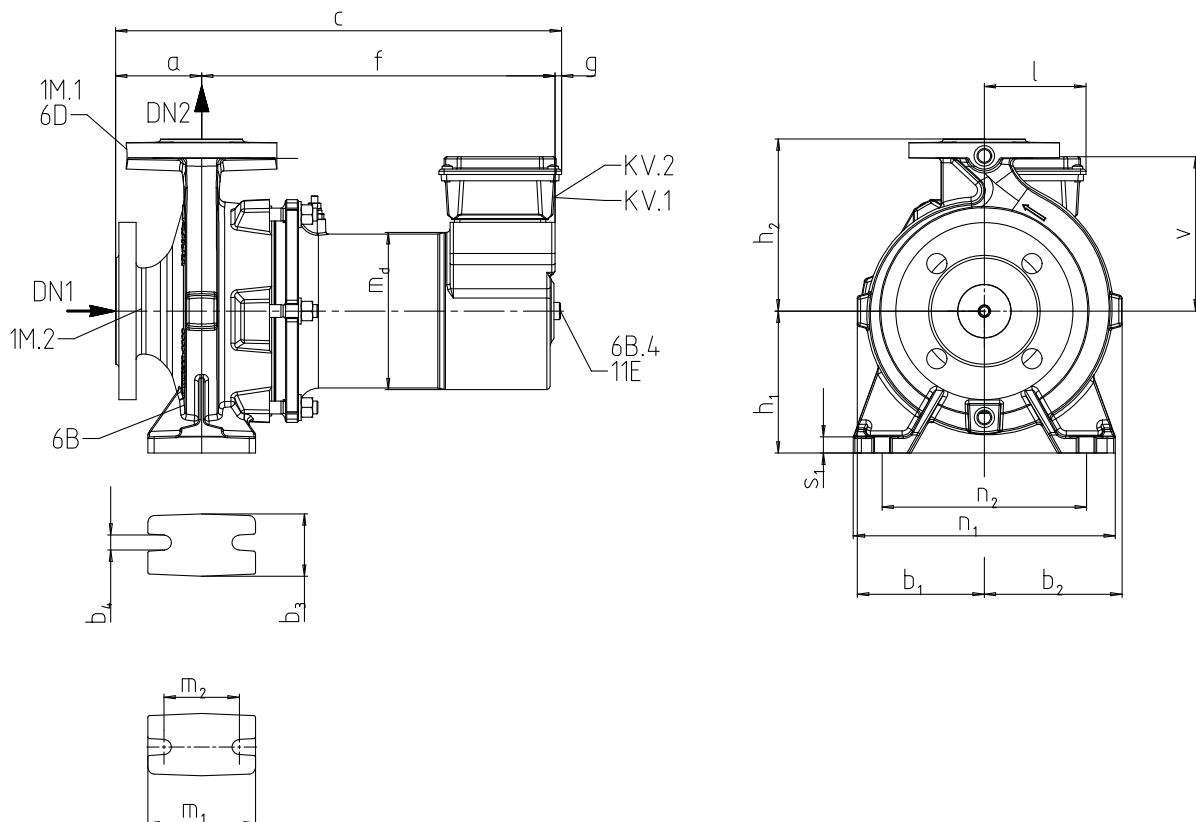


Fig. 1: Dimensions of motor size DS90, motor sizes 22R/S

Table 19: Pump dimensions [mm]

Size	DN1	DN2	a	b ₁	b ₂	b ₃	b ₄	h ₁	h ₂	m ₁	m ₂	n ₁	n ₂	s ₁
040-025-160	040	025	80	118	118	50	14	132	160	100	70	240	190	15
050-032-125	050	032	80	111,5	103	50	14	112	140	100	70	190	140	15
050-032-160	050	032	80	118	128	58	14	132	160	100	70	243,5	190	15
050-032-125.1	050	032	80	108,5	106	50	14	112	140	100	70	190	140	15
050-032-160.1	050	032	80	121	113	50	14	132	160	100	70	240	190	15
065-040-125	065	040	80	117	117	50	14	112	140	100	70	210	160	15

Table 20: Electrical connection dimensions [mm]

Connection type	c	f	g	m _d	v	l
Cable, angular housing on top	409	329	-	146	118,5	-
Cable, angular housing at rear	438	329	31	146	-	-
Terminal box	415	329	6	146	144	95
6B bulkhead housing on top	409	329	-	146	135	79
6B bulkhead housing at rear	455	329	48	146	-	-
3HPR bulkhead housing on top	409	329	-	146	127	84
3HPR bulkhead housing at rear	449	329	42	146	-	-

Connections

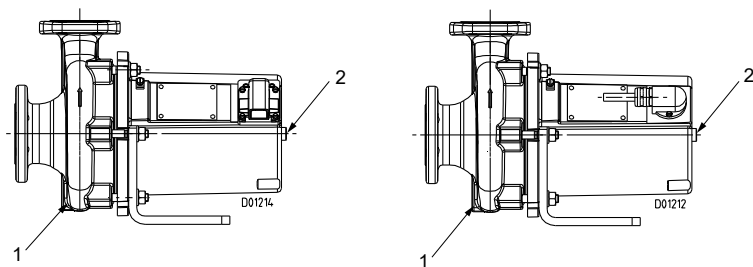


Fig. 2: Connections available for motor size DS90, motor size 12R

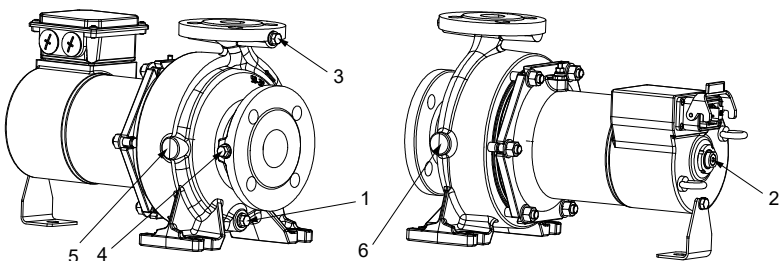


Fig. 3: Connections available for motor size DS90, motor sizes 22R/S

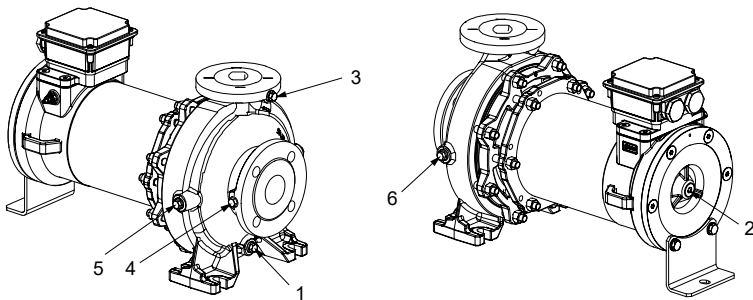


Fig. 4: Connections available for motor sizes DS112 and DS132

Table 21: Connections³⁵⁾

No.	Connection	Description	Motor		
			12R, 22R/S	42S, 52S, 72S, 112S, 152S, 222S	
			Size		
			(all other)		80-65-125/ -160/ -200
1	6B	Drain, pump casing	G 1/4	G 1/4	G 3/8
2	12B	Drain, rotor space	G 1/8	G 3/8	G 3/8
	12D	Vent, rotor space, vertical installation			
3	1M.1	Pressure gauge connection	G 1/4	G 1/4	G 3/8
	6D	Vent, pump casing			
4	1M.2	Pressure gauge connection	G 1/4	G 1/4	G 3/8
5	6B/D.5	See order-specific outline drawing for its function	G 1/4	G 1/4	G 3/8
6	6B/D.6	See order-specific outline drawing for its function	G 1/4	G 1/4	G 3/8

³⁵ If applicable

Flange design

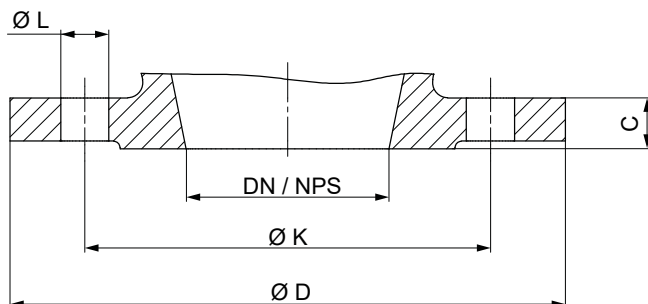


Fig. 5: Flange dimensions

Flange dimensions to EN 1092-1, EN 1092-2

Table 22: Flange dimensions [mm]

Nominal size	Standard							
	EN 1092-1				EN 1092-2			
	Material variant of volute casing							
	C				S/Y			
	PN 10/PN 16				PN 10/PN 16			
	C	Ø D	Ø K	Number x Ø L	C	Ø D	Ø K	Number x Ø L
25	19	115	85	4 × Ø 14	15	115	85	4 × Ø 14
32	19	140	100	4 × Ø 18	17	140	100	4 × Ø 19
40	19	150	110	4 × Ø 18	17	150	110	4 × Ø 19
50	19	165	125	4 × Ø 18	19	165	125	4 × Ø 19
65	19	185	145	4 × Ø 18	19	185	145	4 × Ø 19
80 (DN ₁ ³⁶)	21	230	160	8 × Ø 18	21	229	160	8 × Ø 19

Flange drilled to ASME B16.5, Class 150

Table 23: Flange dimensions [mm]

Nominal size	Standard							
	ASME B16.5, Class 150							
	Material variant of volute casing							
	C				S/Y			
	C	Ø D	Ø K	Number x Ø L	C	Ø D	Ø K	Number x Ø L
	C	Ø D	Ø K	Number x Ø L	C	Ø D	Ø K	Number x Ø L
25/NPS 1	19	115	79,2	4 x Ø 15,9	15	115	79	4 x Ø 15,7
32/NPS 1 1/4	19	140	88,9	4 x Ø 15,9	17	140	89	4 x Ø 15,7
40/NPS 1 1/2	19	150	98,6	4 x Ø 15,9	17	150	98	4 x Ø 15,7
50/NPS 2	19	165	120,7	4 x Ø 19,1	19	165	121	4 x Ø 19,1
65/NPS 2 1/2	19	185	139,7	4 x Ø 19,1	19	185	140	4 x Ø 19,1
80/NPS 3 (DN ₁ ³⁶⁾	21	230	190,5	8 x Ø 19,1	21	229	191	8 x Ø 19,1

Flange drilled to SAE ISO 6162-1
Table 24: Flange dimensions [mm]

Nominal size		Standard									
		ISO 6162-1									
		Material variant of volute casing									
		C					S/Y				
[DN]	[inch]	C ³⁷⁾	Ø D ³⁸⁾	K1	K2	Number x Ø L	C ³⁷⁾	Ø D ³⁸⁾	K1	K2	Number x Ø L
25	1	14,7	115	52,4	26,2	4 × M 10	11,7	115	52,4	26,2	4 × M 10
51	2	15,2	150	77,8	42,9	4 × M 12	16	150	77,8	42,9	4 × M 12
76	3	15	185	106,4	61,9	4 × M 16	16	185	106,4	61,9	4 × M 16
89	3 1/2	16,8	200	120,7	61,9	4 × M 16	22	200	120,7	69,9	4 × M 16

SAE flange dimensions to ISO 6162-1³⁹⁾
Table 25: Flange dimensions [mm]

Nominal size		Standard					
		ISO 6162-1					
		Material variant of volute casing					
		S/Y					
[DN]	[inch]	C	D1	D2	K1	K2	Number x Ø L
51	2	15	105	100	77,8	42,9	4 × M 12

37 The flange thickness may deviate slightly. Compliance with the pressure class (PN 10/PN 16) is ensured.

38 The outside diameter may deviate slightly.

39 For sizes 50-32-125.0 and 50-32-125.2 only

General assembly drawing with list of components

General assembly drawing with list of components, motor size DS90, motor size 12R

Version with bulkhead housing

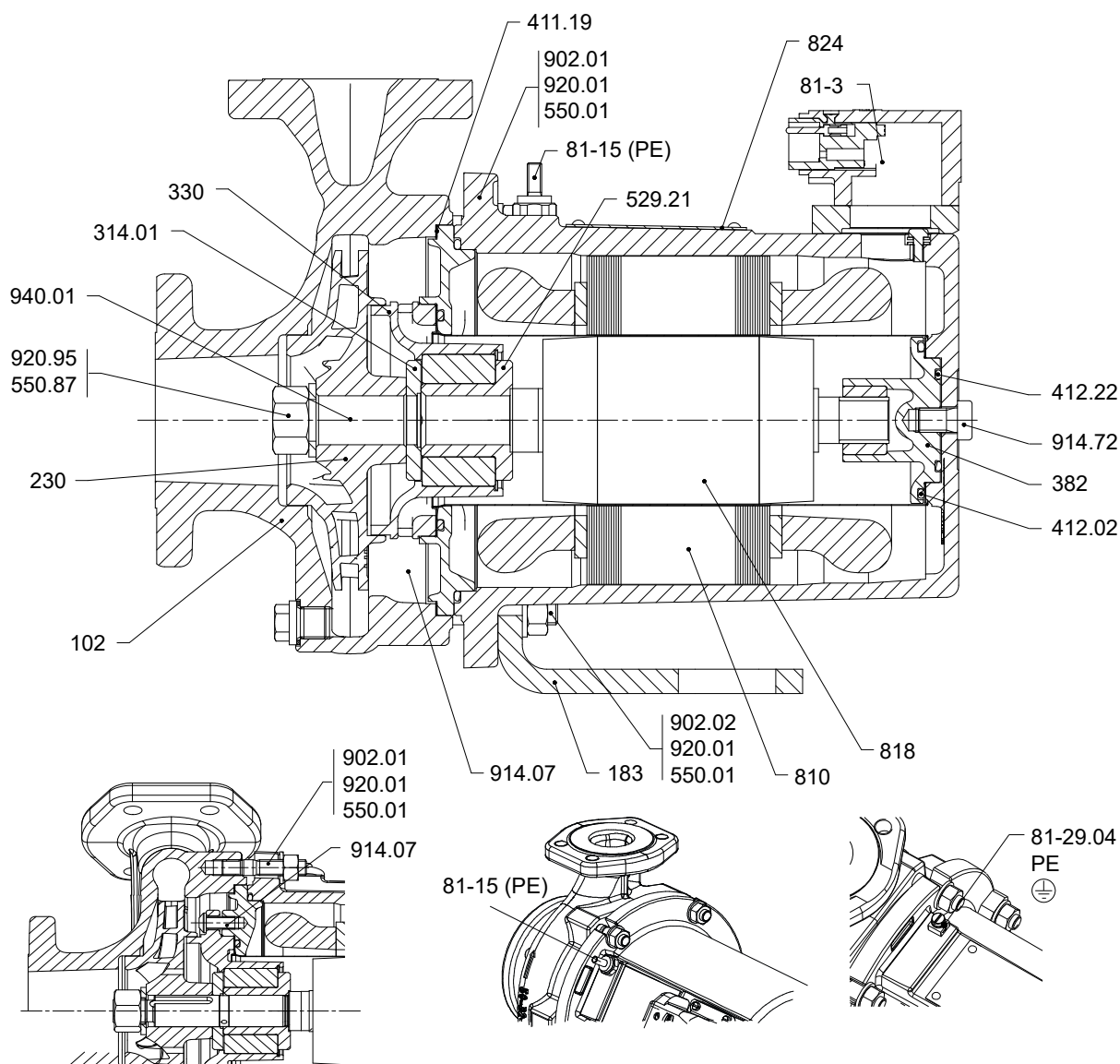


Fig. 6: General assembly drawing of version with bulkhead housing

Table 26: List of components

Part No.	Part No.	Description
102	102	Volute casing
	902.01/.02	Stud
	550.01	Disc
	920.01	Hexagon nut
183	183	Angle support foot
230	230	Impeller
	-	Plain bearing
	314.01	Thrust bearing
	330	Bearing bracket
	529.21	Bearing sleeve
	914.07	Hexagon socket head cap screw
314.01	314.01	Thrust bearing
330	330	Bearing bracket

Part No.	Part No.	Description
330	-	Bearing bush
	-	Circlip
	914.07	Hexagon socket head cap screw
382	382	Bearing carrier
	-	Bearing bush
	914.72	Hexagon socket head cap screw
529.21	529.21	Bearing sleeve
81-11	81-11	PE terminal stud (assembly)
	81-15.58	PE terminal stud
	920.58	Hexagon nut
	930.58	Detent-edged washer
810	810	Motor part
	-	Motor housing
	-	Stator
	-	Motor housing cover / discharge cover
	-	Can
	81-11	Potential equalisation connection
	81-29.04	Optional: potential equalisation terminal
818	818	Rotor
	-	Bearing sleeve (motor end)
	940.01	Key
920.95	920.95	Hexagon nut
	550.87	Disc
99-9	99-9	Set of sealing elements
	411.19	Gasket
	412.02/.22	O-ring
91-99	91-99	Electrical connection
	81-99.01/.02	Bulkhead housing

Version with integrated power cable

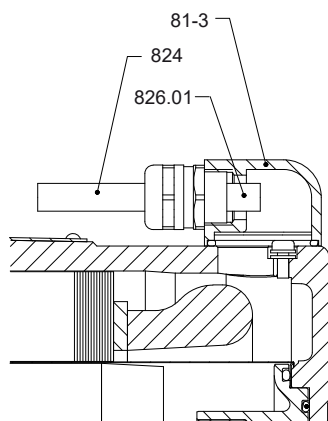


Fig. 7: Detail drawing, connection via integrated power cable

Table 27: List of components: integrated power cable

Part No.	Part No.	Description
-	-	Electrical connection
	824	Integrated power cable
	826.01	Cable gland
	81-3	Bushing

General assembly drawing with list of components, motor size DS90, motor sizes 22R/S

Carbon/stainless steel plain bearing version

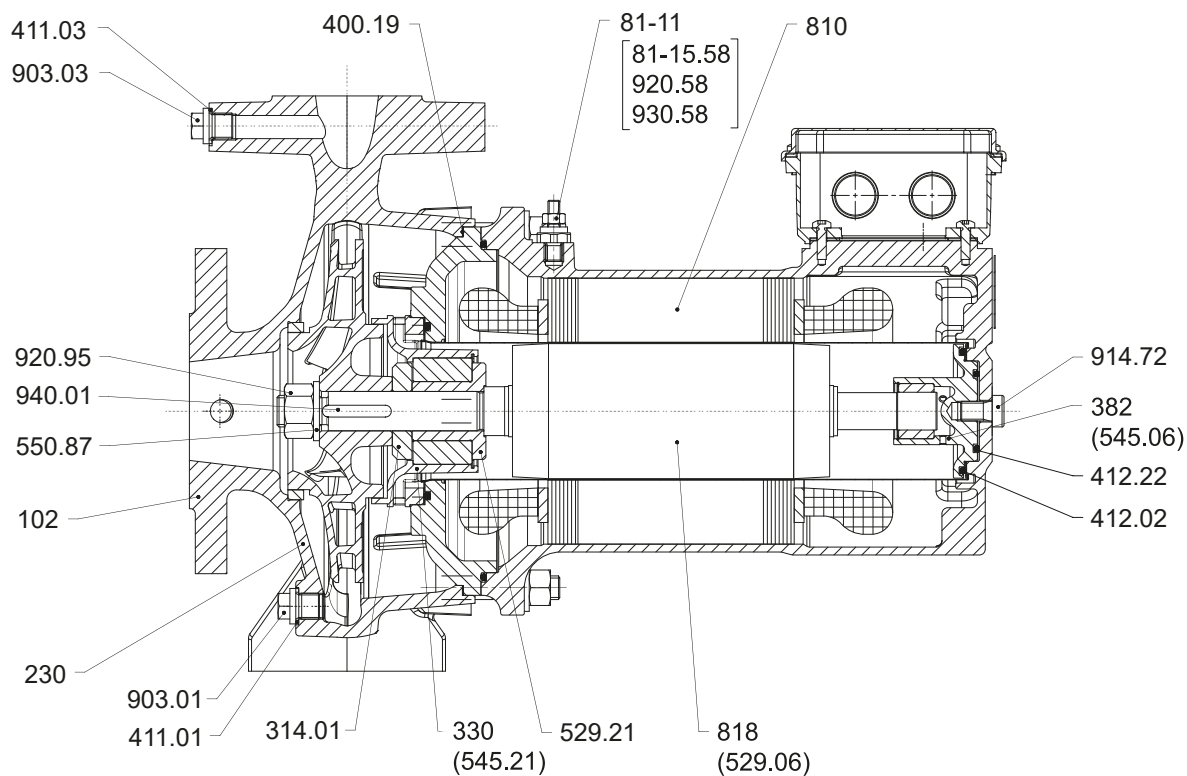


Fig. 8: General assembly drawing: Pump set with carbon/stainless steel plain bearings

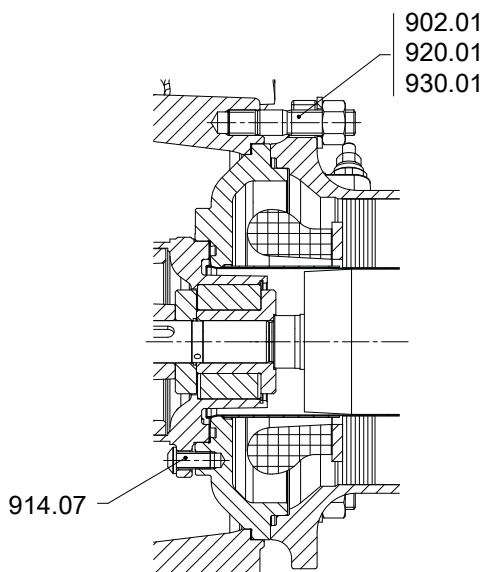


Fig. 9: Detail drawing: connection between pump and motor

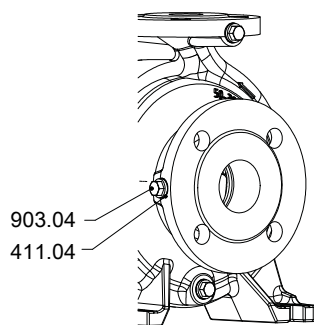


Fig. 10: Detail drawing: screw plug at suction-side flange

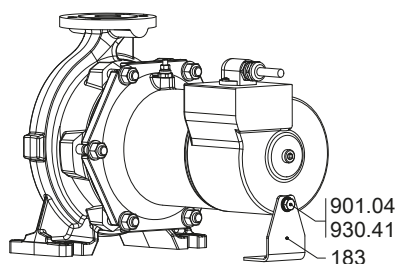


Fig. 11: Detail drawing: support foot

Table 28: List of components

Part No.	Comprising	Description
102	102	Volute casing
	400.19	Gasket (casing gasket)
	411.01/.03 ⁴⁰⁾ /.04 ⁴⁰⁾	Joint ring
	-	Casing wear ring (502.01) ⁴¹⁾⁴²⁾
	902.01	Stud
	903.01/.03 ⁴⁰⁾ /.04 ⁴⁰⁾	Screw plug
	920.01	Hexagon nut
	930.01	Detent-edged washer
183 ⁴¹⁾	183	Support foot
	901.04	Hexagon head bolt
	930.41	Detent-edged washer
230	-	Impeller
314.01	-	Thrust bearing
330	300	Bearing bracket
	914.07	Hexagon socket head cap screw
	-	Bearing bush (545.21)
382	382	Bearing carrier without 12D (assembly)
	914.72	Hexagon socket head cap screw
	-	Bearing bush (545.06)
382	382	Bearing carrier with 12D (assembly)
	411.91	Joint ring
	903.91	Screw plug
	920.72	Shaft nut
	-	Bearing bush (545.06)
400.19	-	Gasket
411.01/.03/.04/.91	-	Joint ring
412.02/.22	-	O-ring
529.06	-	Bearing sleeve, drive-end (see rotor 818)
529.21	-	Bearing sleeve, pump-end
550.87	-	Disc

40 Optional

41 If applicable

42 Not intended to be replaced

Part No.	Comprising	Description
81-11	81-11	PE terminal stud (assembly)
	81-15.58	PE terminal stud
	920.58	Hexagon nut
	930.58	Detent-edged washer
810	810	Motor part
	412.21/.22	O-ring
	81-11	PE terminal stud
	-	Motor housing
	-	Stator
	-	Can
	-	Discharge cover
818	818	Rotor
	940.01	Key
	-	Bearing sleeve, drive end (529.06)
900.82 ⁴¹⁾	-	Eyebolt
920.95	-	Impeller nut

SiC/SiC plain bearing version

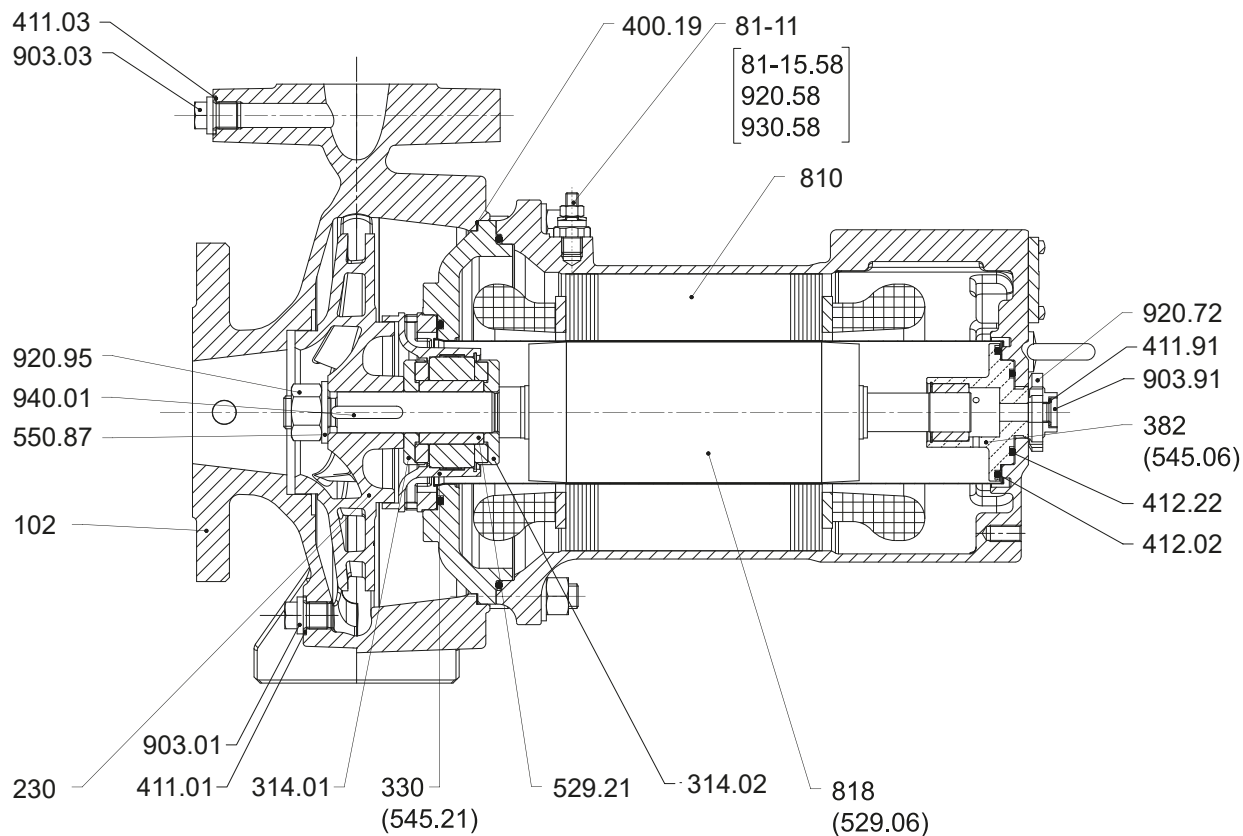


Fig. 12: General assembly drawing: Pump set with SiC/SiC plain bearings

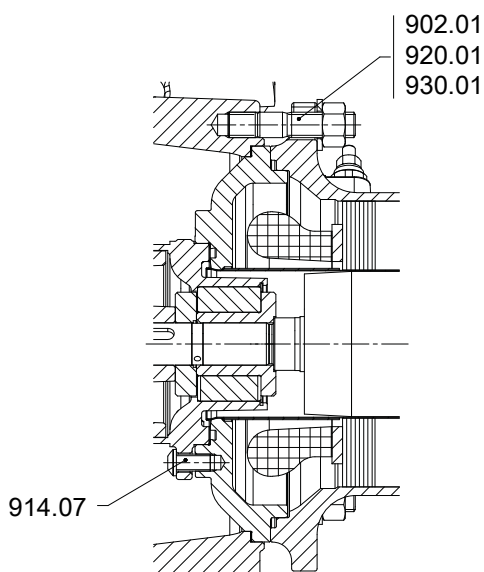


Fig. 13: Detail drawing: connection between pump and motor

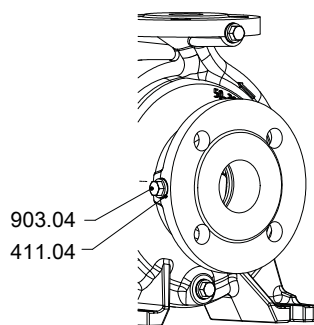


Fig. 14: Detail drawing: screw plug at suction-side flange

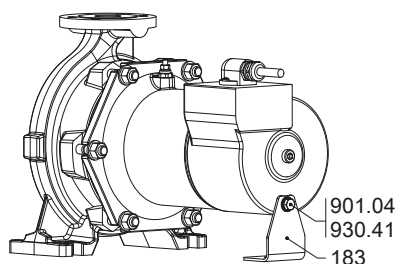


Fig. 15: Detail drawing: support foot

Table 29: List of components

Part No.	Comprising	Description
102	102	Volute casing
	400.19	Gasket (casing gasket)
	411.01/.03 ⁴³⁾ /.04 ⁴³⁾	Joint ring
	-	Casing wear ring (502.01) ⁴⁴⁾ ⁴⁵⁾
	902.01	Stud
	903.01/.03 ⁴³⁾ /.04 ⁴³⁾	Screw plug
	920.01	Hexagon nut
	930.01	Detent-edged washer
183 ⁴⁴⁾	183	Support foot
	901.04	Hexagon head bolt
	930.41	Detent-edged washer
230	-	Impeller
314.01/.02	-	Thrust bearing
330	330	Bearing bracket
	914.07	Hexagon socket head cap screw
	-	Bearing bush (545.21)
382	382	Bearing carrier without 12D (assembly)
	914.72	Hexagon socket head cap screw
	-	Bearing bush (545.06)
382	382	Bearing carrier with 12D (assembly)
	411.91	Joint ring
	903.91	Screw plug
	920.72	Shaft nut
	-	Bearing bush (545.06)
400.19	-	Gasket
411.01/.91	-	Joint ring
412.02/.22	-	O-ring
529.06	-	Bearing sleeve, drive end (see rotor 818)
529.21	-	Bearing sleeve, pump-end
550.87	-	Disc

43 Optional

44 If applicable

45 Not intended to be replaced

Part No.	Comprising	Description
810	810	Motor part
	412.21/.22	O-ring
	81-11	PE terminal stud
	-	Motor housing
	-	Stator
	-	Can
	-	Discharge cover
81-11	81-11	PE terminal stud (assembly)
	81-15.58	PE terminal stud
	920.58	Hexagon nut
	930.58	Detent-edged washer
818	818	Rotor
	940.01	Key
	-	Bearing sleeve, drive end (529.06)
900.82 ⁴⁴⁾	-	Eyebolt
920.95	-	Impeller nut

Electrical connections

Connection via bulkhead housing

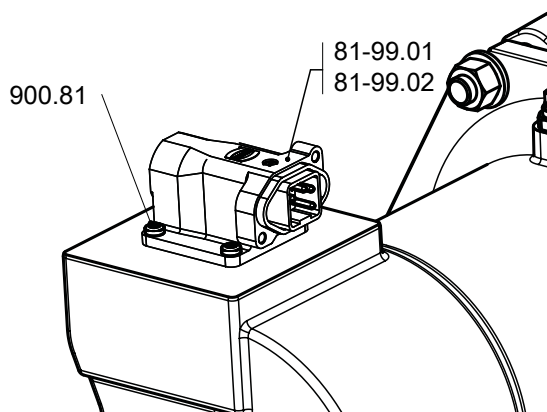


Fig. 16: Detail drawing: 3HPR bulkhead housing, top position

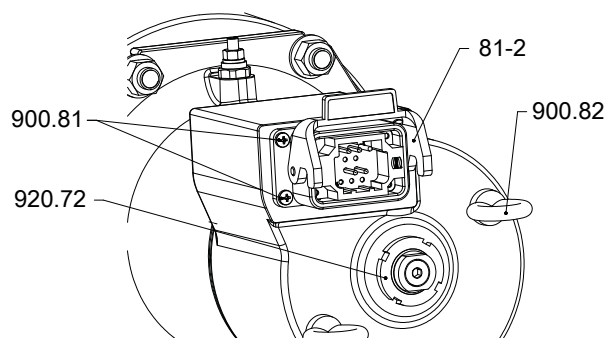


Fig. 17: Detail drawing: 6B bulkhead housing, rear position

Connection via integrated power cable

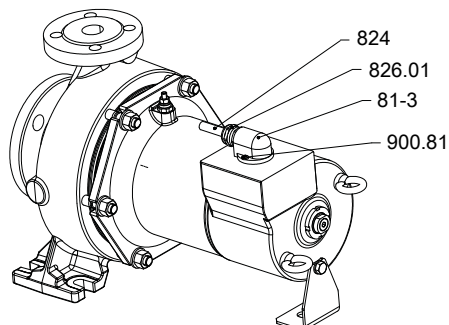


Fig. 18: Detail drawing: integrated power cable, top position

Connection via terminal box

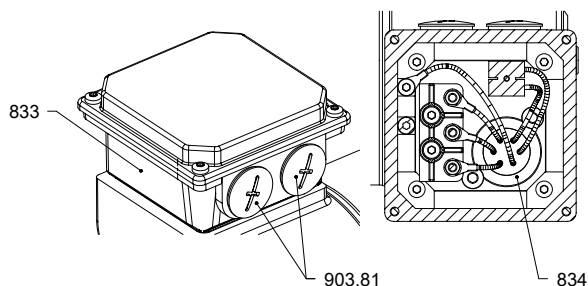


Fig. 19: Detail drawing: terminal box

Table 30: List of components, electrical connections

Part No.	Comprising	Description
Electrical connection		
-	-	Power cable
-	824	Electric cable
-	826.01	Cable gland
-	81-3 ⁴⁶⁾	Flanged angle gland with O-ring
-	900.81 ⁴⁶⁾	Bolt/screw
-	-	Bulkhead housing 3HPR
-	81-99.01	Bulkhead housing with sealing element
-	81-99.02	Pin insert with contact pins
-	900.81	Bolt/screw
-	-	Bulkhead housing 6B
-	81-99.01	Bulkhead housing with sealing element
-	81-99.02	Pin insert with contact pins
-	900.81	Bolt/screw
-	-	Terminal box
-	833	Terminal box with terminal block, cover, and sealing element
-	834	Sealed terminal gland or cable gland
-	903.81	Screw plugs on terminal box

Vent connection (12D) of the rotor space

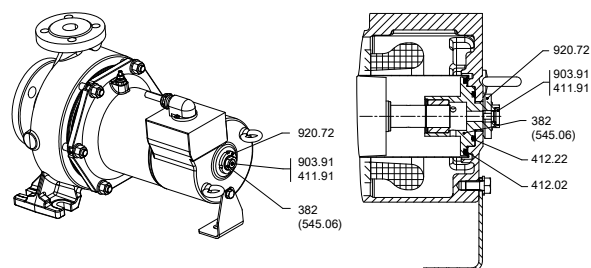


Fig. 20: Vent connection (12D)

Table 31: List of components

Part No.	Description
382	Bearing carrier with bearing bush (545.06)
411.91	Joint ring
412.02/22	O-ring
903.91	Screw plug
920.72	Shaft nut
-	Bearing bush (545.06)

General assembly drawing with list of components, motor sizes DS112 and DS132

Carbon/stainless steel plain bearing version

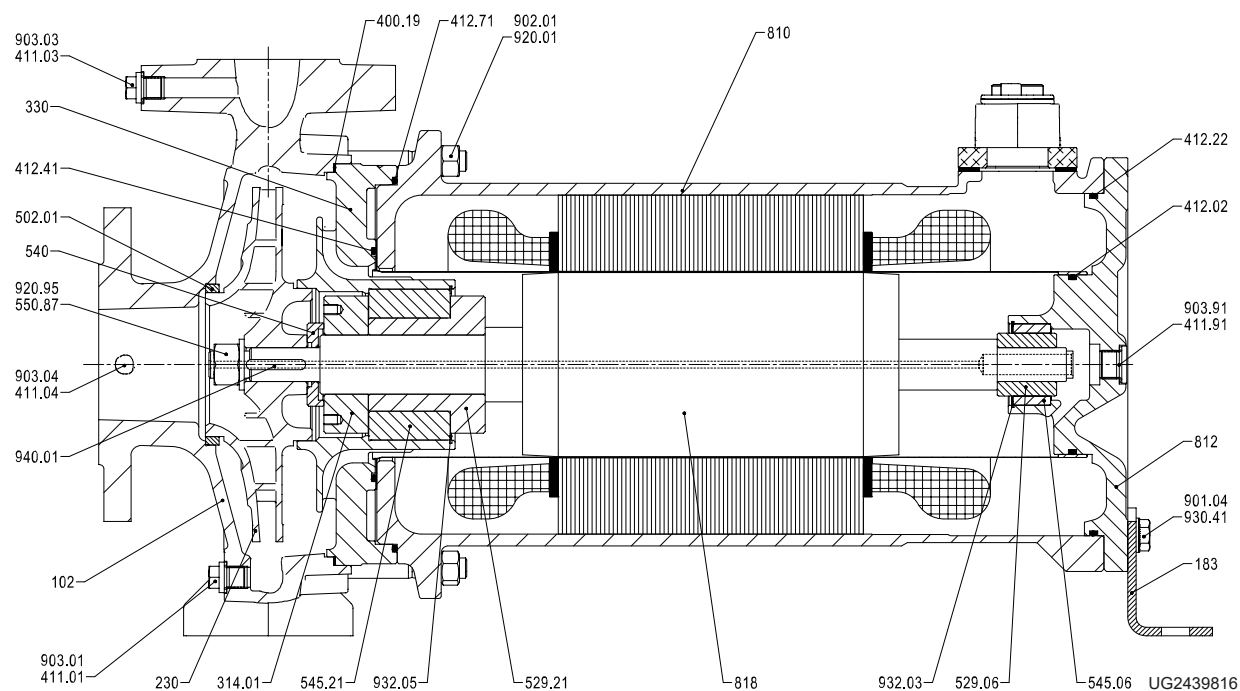


Fig. 21: General assembly drawing: Pump set with carbon/stainless steel plain bearings (version with clamped bearing bracket)

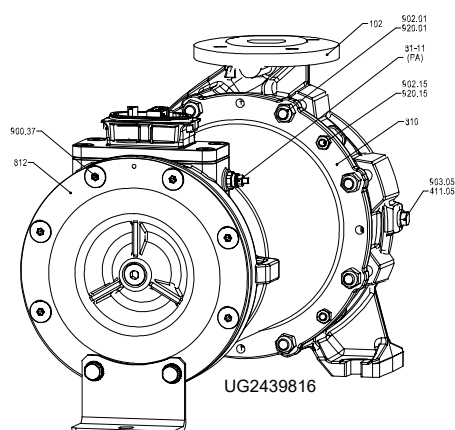


Fig. 22: Detail drawing, PE terminal stud and screwed/bolted connections

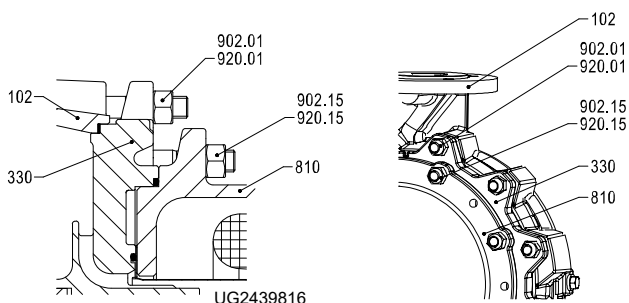


Fig. 23: Version with bolted bearing bracket for carbon/stainless steel plain bearings

Table 32: List of components

Part No.	Comprising	Description
102	102	Volute casing (pump casing)

Part No.	Comprising	Description
102	411.01/.03/.04/.05/.06 ⁴⁷⁾	Joint ring
	-	Casing wear ring (502.01) ⁴⁷⁾⁴⁸⁾
	902.01	Stud
	903.01/.03/.04/.05/.06 ⁴⁷⁾	Screw plug
	920.01	Nut
183 ⁴⁷⁾	183	Support foot
	901.04	Hexagon head bolt
	930.41	Detent-edged washer
230	230	Impeller
- ⁴⁹⁾	-	Plain bearing, pump end (assembly)
	314.01	Thrust bearing
	529.21	Bearing sleeve
330	330	Bearing bracket
	400.19	Gasket
	412.41	O-ring
	545.21	Bearing bush
	902.15	Stud
	920.15	Nut
	932.05	Circlip
540	540 ⁴⁷⁾	Bush
550.87	550.87	Disc
81-11	81-11	PE terminal stud (assembly)
810	810	Motor part (assembly)
	412.71	O-ring
	-	Motor housing
	-	Stator
	-	Can
	-	Support
	-	Plate for the power supply connection
	-	Gasket
	-	Countersunk head screw
812	812	Motor housing cover
	411.91	Joint ring
	412.02	O-ring
	412.22	O-ring
	545.06	Bearing bush (motor end)
	900.37	Countersunk head screw
	903.91	Screw plug
	932.03	Circlip
818	818	Rotor
	529.06	Bearing sleeve (motor end), cannot be removed
	940.01	Key
	-	Shaft
920.95	920.95	Nut
99-9	99-9	Set of sealing elements
	400.19	Gasket
	411.01/.03/.04/.05/.06/.91 ⁴⁷⁾	Joint ring
	412.02/.22/.41/.71 ⁴⁷⁾	O-ring

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- 47** If applicable
48 Not intended to be replaced
49 Rotating components only

SiC/SiC and hybrid plain bearings

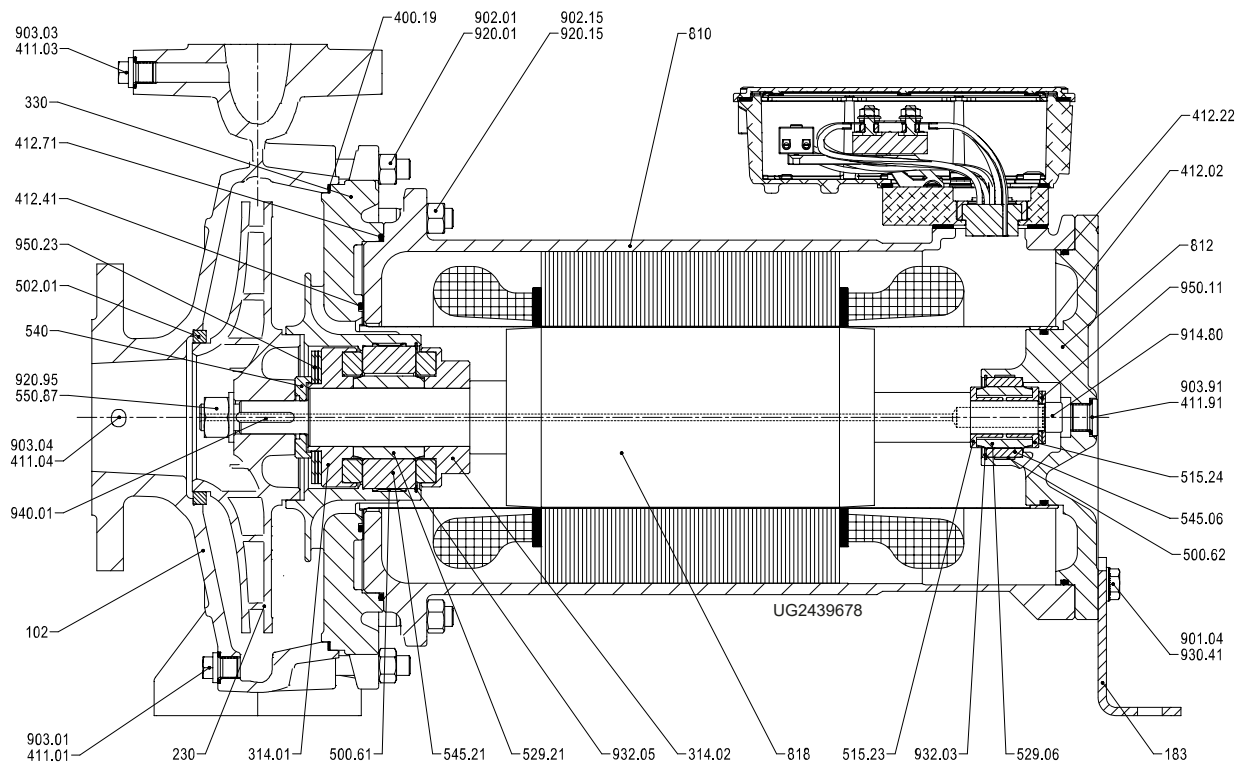


Fig. 24: General assembly drawing: Pump set with SiC/SiC and hybrid plain bearings (version with bolted bearing bracket)

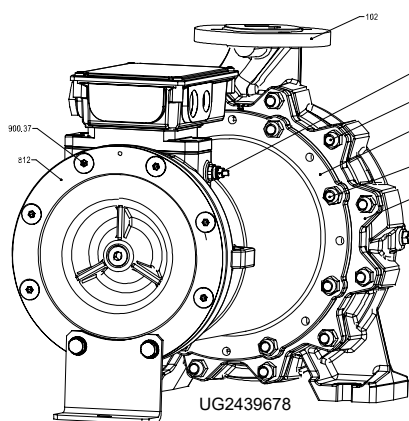


Fig. 25: Detail drawing, PE terminal stud and screwed/bolted connections

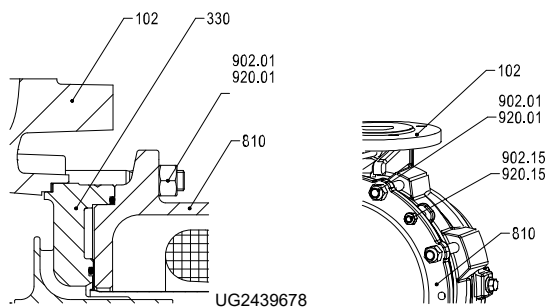
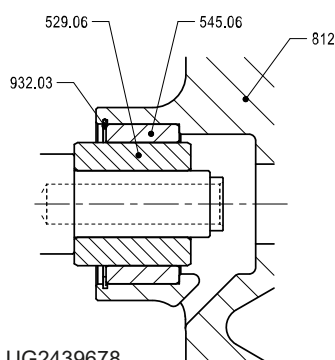


Fig. 26: Clamped bearing bracket version



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Fig. 27: Motor-end plain bearing for version with hybrid plain bearings (pump-end SiC/SiC plain bearing and motor-end carbon/stainless steel plain bearing)

Table 33: List of components

Part No.	Comprising	Description
102	102	Volute casing (pump casing)
	411.01/.03/.04/.05/.06 ⁵⁰⁾	Joint ring
	-	Casing wear ring (502.01) ⁵⁰⁾⁵¹⁾
	902.01	Stud
	903.01/.03/.04/.05/.06 ⁵⁰⁾	Screw plug
	920.01	Nut
183 ⁵⁰⁾	183	Support foot
	901.04	Hexagon head bolt
	930.41	Detent-edged washer
230	230	Impeller
_52)	-	Plain bearing, pump end (assembly)
	314.01/.02	Thrust bearing
	529.21	Bearing sleeve (pump end)
	950.23	Disc spring
_52)	-	Plain bearing, motor end (assembly)
	515.23/.24 ⁵³⁾	Locking ring
	529.06	Bearing sleeve (motor end)
	914.80 ⁵³⁾	Hexagon socket head cap screw (left-hand thread)
	950.11 ⁵³⁾	Disc springs
330	330	Bearing bracket
	400.19	Gasket
	412.41	O-ring
	500.61	Tolerance ring
	545.21	Bearing bush (pump end)
	902.15	Stud
	920.15	Nut
	932.05	Circlip
540	540	Bush
550.87	550.87	Disc
81-11	81-11	PE terminal stud (assembly)
810	810	Motor part (assembly)
	412.71	O-ring
	-	Motor housing
	-	Stator
	-	Can
	-	Support
	-	Plate for the power supply connection

50 If applicable

51 Not intended to be replaced

52 Rotating components only

53 Not available for hybrid plain bearings

Part No.	Comprising	Description
810	-	Gasket
812	812	Motor housing cover
	411.91	Joint ring
	412.02	O-ring
	412.22	O-ring
	500.62 ⁵³⁾	Tolerance ring
	545.06	Bearing bush (motor end)
	900.37	Countersunk head screw
	903.91	Screw plug
	932.03	Circlip
818	818	Rotor
	940.01	Key
	-	Shaft
920.95	920.95	Nut
99-9	99-9	Set of sealing elements
	400.19	Gasket
	411.01/.03/.04/.05/.06/.91 ⁵⁰⁾	Joint ring
	412.02/.22/.41/.71 ⁵⁰⁾	O-ring

Electrical connection

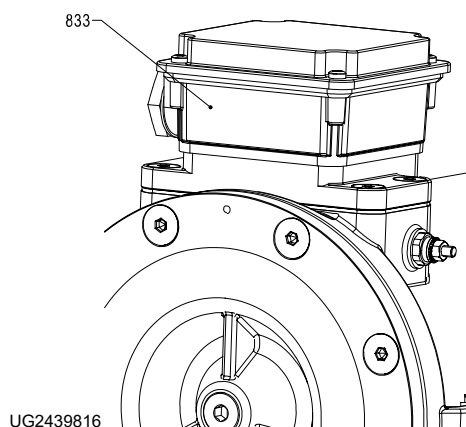


Fig. 28: Detail drawing, terminal box for motor size DS112

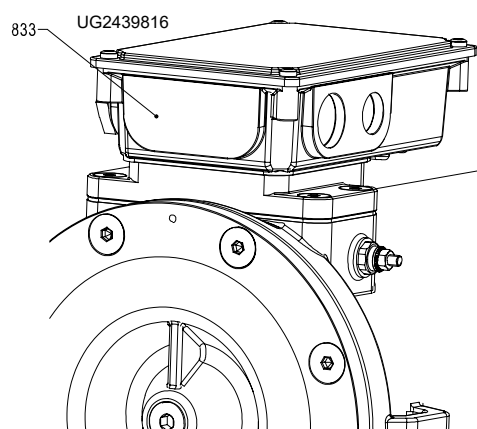


Fig. 29: Detail drawing, terminal box for motor size DS132

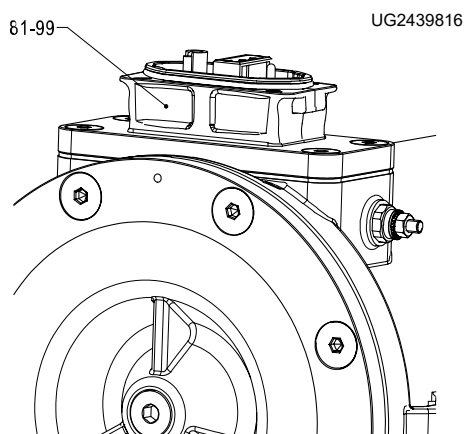


Fig. 30: Detail drawing, bulkhead housing (example: 6HPR)

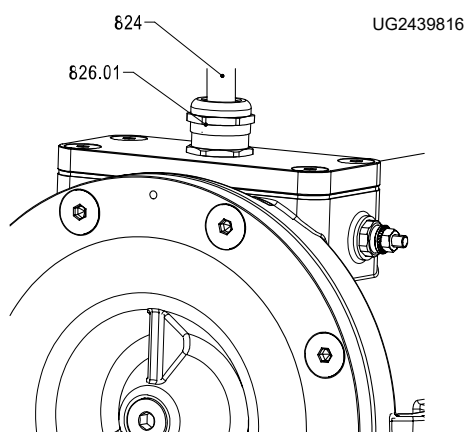


Fig. 31: Detail drawing, integrated power cable

Table 34: List of components

Part No.	Comprising	Description
Electrical connection for version with terminal box ⁵⁴⁾		
833	833	Terminal box
Electrical connection for version with bulkhead housing ⁵⁴⁾		
81-99	81-99	Bulkhead housing (3HPR/6B/6HPR/10HPR/16HPR) ⁵⁵⁾
Electrical connection for version with integrated power cable ⁵⁴⁾		
-	-	Power cable
	824	Electric cable
	826.01	Cable gland

54 If applicable

55 Depending on the version



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